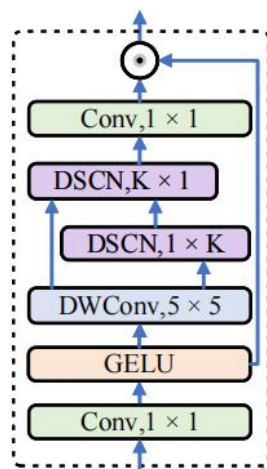


论文信息



(b)

图 5. DSA, 由一对 DSCN 操作、两个作用于通道域的线性操作和一个 GELU 激活函数组成

论文题目： DSAN: Exploring the Relationship between Deformable Convolution and Spatial Attention

中文题目： DSAN: 探索可变形卷积与空间注意力之间的关系

论文链接：

https://d197for5662m48.cloudfront.net/documents/publicationstatus/204101/preprint_pdf/8fd08fb8f6c83b6c5ad1eec37e87faf0.pdf

官方 github： <https://github.com/MarcYugo/DSAN-Deformable-Spatial-Attention>

所属机构：温州大学等

核心速览：本文提出了一种名为 DSAN (Deformable Spatial Attention Network) 的轻量级卷积神经网络 (CNN) 架构，旨在解决现有变形卷积神经网络 (DCNv3) 在轻量级模型中应用时存在的性能不足问题。

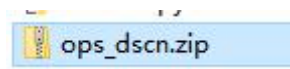
Deformable Strip Convolution (DSCN): 作者提出了一种简化版的 DCNv3 核心操作，称为 DSCN。DSCN 通过限制变形采样核仅在单一轴上进行采样，避免了因核大小增加而导致的计算负载和参数数量的二次增长。与 DCNv3 的核心操作相比，DSCN 在计算负载和参数数量上都有显著减少。

Deformable Spatial Attention (DSA): 基于 DSCN，作者进一步提出了 DSA 作为 DCNv3 的替代方案。DSA 通过使用空间注意力机制替代 DCNv3 中的调制掩码分支，从而减少了参数和内存消耗。

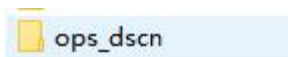
使用步骤

步骤 1: 安装 DSCN (注意安装过程均在 pytorch 虚拟环境中操作)

①把安装程序压缩包放在自己想放置的目录下, 然后进行解压;



②解压后, 在命令行输入 cd ops_dscn 进入该文件夹;



③进入该目录后, 执行命令 python setup.py install 进行安装;

```
正在创建库 build\temp.win-amd64-cpython-39\Release\AI\External-Attention-pytorch-master\ops_dscn\src\cpu\DSCN.cp39-win_amd64.lib 和对象 build\temp.win-amd64-cpython-39\Release\AI\Ext
pytorch-master\ops_dscn\src\cpu\DSCN.cp39-win_amd64.exp
正在生成代码
已完成代码的生成
creating build\bdist.win-amd64
creating build\bdist.win-amd64\egg
copying build\lib.win-amd64-cpython-39\DSCN.cp39-win_amd64.pyd -> build\bdist.win-amd64\egg
creating build\bdist.win-amd64\egg\functions
copying build\lib.win-amd64-cpython-39\functions\dscn_func.py -> build\bdist.win-amd64\egg\functions
copying build\lib.win-amd64-cpython-39\functions\__init__.py -> build\bdist.win-amd64\egg\functions
creating build\bdist.win-amd64\egg\modules
copying build\lib.win-amd64-cpython-39\modules\dscn.py -> build\bdist.win-amd64\egg\modules
copying build\lib.win-amd64-cpython-39\modules\__init__.py -> build\bdist.win-amd64\egg\modules
byte-compiling build\bdist.win-amd64\egg\functions\dscn_func.py to dscn_func.cpython-39.pyc
byte-compiling build\bdist.win-amd64\egg\functions\__init__.py to __init__.cpython-39.pyc
byte-compiling build\bdist.win-amd64\egg\modules\dscn.py to dscn.cpython-39.pyc
byte-compiling build\bdist.win-amd64\egg\modules\__init__.py to __init__.cpython-39.pyc
creating stub loader for DSCN.cp39-win_amd64.pyd
byte-compiling build\bdist.win-amd64\egg\DSCN.py to DSCN.cpython-39.pyc
creating build\bdist.win-amd64\egg\EGG-INFO
copying DSCN.egg-info\PKG-INFO -> build\bdist.win-amd64\egg\EGG-INFO
copying DSCN.egg-info\SOURCES.txt -> build\bdist.win-amd64\egg\EGG-INFO
copying DSCN.egg-info\dependency_links.txt -> build\bdist.win-amd64\egg\EGG-INFO
copying DSCN.egg-info\top_level.txt -> build\bdist.win-amd64\egg\EGG-INFO
writing build\bdist.win-amd64\egg\EGG-INFO\native_libs.txt
zip_safe flag not set; analyzing archive contents...
__pycache__.DSCN.cpython-39: module references __file__
creating dist
creating 'dist\DSCN-0.1-py3.9-win-amd64.egg' and adding 'build\bdist.win-amd64\egg' to it
removing 'build\bdist.win-amd64\egg' (and everything under it)
Processing DSCN-0.1-py3.9-win-amd64.egg
creating d:\anaconda\envs\learning1\lib\site-packages\DSCN-0.1-py3.9-win-amd64.egg
Extracting DSCN-0.1-py3.9-win-amd64.egg to d:\anaconda\envs\learning1\lib\site-packages
Adding DSCN 0.1 to easy-install.pth file

Installed d:\anaconda\envs\learning1\lib\site-packages\dscn-0.1-py3.9-win-amd64.egg
Processing dependencies for DSCN==0.1
Finished processing dependencies for DSCN==0.1
```

④会出现一大堆安装提示, 等待即可, 出现上面信息, 表示安装成功!!!

步骤 2: 运行测试代码

执行(TNNLS 2025) Deformable Spatial Attention (DSA).py

结果如下: (注意上面py程序和ops_dscn保持同级目录)

```
DSA(  
  (proj_1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1))  
  (activation): GELU(approximate='none')  
  (spatial_gating_unit): DSCNPair(  
    (conv0): Conv2d(64, 64, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=64)  
    (dscn_x): DSCNX(  
      (dw_conv): Sequential(  
        (0): Conv2d(64, 64, kernel_size=(1, 5), stride=(1, 1), padding=(0, 2), groups=64)  
        (1): Sequential(  
          (0): to_channels_last()  
          (1): LayerNorm((64,)), eps=1e-06, elementwise_affine=True)  
        )  
        (2): GELU(approximate='none')  
      )  
      (offset): Linear(in_features=64, out_features=11, bias=True)  
      (input_proj): Linear(in_features=64, out_features=64, bias=True)  
    )  
    (dscn_y): DSCNY(  
      (dw_conv): Sequential(  
        (0): Conv2d(64, 64, kernel_size=(5, 1), stride=(1, 1), padding=(2, 0), groups=64)  
        (1): Sequential(  
          (0): to_channels_last()  
          (1): LayerNorm((64,)), eps=1e-06, elementwise_affine=True)  
        )  
        (2): GELU(approximate='none')  
      )  
      (offset): Linear(in_features=64, out_features=11, bias=True)  
    )  
    (conv): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1))  
  )  
  (proj_2): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1))  
)  
微信公众号: AI缝合术  
输入张量形状: torch.Size([1, 64, 128, 128])  
输出张量形状: torch.Size([1, 64, 128, 128])
```